

Solutions for Mathematical Proofs: A Transition
to Advanced Mathematics

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Chapter 3

Direct Proof and Proof by Contrapositive

Result 3.1. *Prove that if x is an odd integer, then $9x + 5$ is even.*

Proof. Let $x = 2k + 1$ such that $k \in \mathbb{Z}$

$$\begin{aligned} 9x + 5 &= 9(2k + 1) + 5 \\ &= 18k + 14 \\ &= 2(9k + 7) \end{aligned}$$

, since $9k + 7$ is an integer, thus $9x + 5$ is even □

Result 3.2. *Prove that if x is an even integer, then $5x - 3$ is an odd integer*

Proof. Let $x = 2k$, such that $k \in \mathbb{Z}$

$$\begin{aligned} 5x - 3 &= 5(2k) - 3 \\ &= 10k - 3 \\ &= (10k - 2) - 1 \\ &= 2(5k - 1) - 1 \end{aligned}$$

since $5k - 1$ is an integer, then $5x - 3$ is an odd integer □

Result 3.3. *Let $n \in \mathbb{Z}$ Prove that $1 - n^2 > 0$ then $3n - 2$ is an even integer*

Proof. Proving by contrapositive of the implication, $1 - n^2 > 0$ then $3n - 2$ is an even integer, we have $3n - 2$ is an odd number gives $1 - n^2 \leq 0$. let $n = 2k + 1$ then we have

$$\begin{aligned} 1 - n^2 &= 1 - (2k + 1)^2 \\ &= 1 - 4k^2 - 4k - 1 \\ &= -4k^2 - 4k \end{aligned}$$

which $-4k^2 - 4k$ is an integer, always less than or equal to 0 □

Result 3.4. *Let $x \in \mathbb{Z}$ Prove that if $7x + 5$ is odd, then x is even*

Proof. Proving by contrapositive of the implication, if $7x + 5$ is odd, then x is even, is if x is odd, then $7x + 5$ is even. Let $x = 2k + 1$ such that $k \in \mathbb{Z}$.

$$\begin{aligned} 7x + 5 &= 7(2k + 1) + 5 \\ &= 14k + 7 + 5 = 14k + 12 \\ &= 2(7k + 6) \end{aligned}$$

which $2(7k + 6)$ is an even integer □